

Cristian F. López R.

(he/him/his)

September, 2026

Postdoctoral Associate

Department of Applied Mathematics
University of Colorado Boulder

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EDUCATION

2021 – 2025	Ph.D. in Mechanical Engineering and Applied Mechanics University of Nebraska – Lincoln, Lincoln, Nebraska, USA Dissertation: “Data-driven Methods for Signal Analysis and Nonlinear System Identification” M.S. in Mechanical Engineering and Applied Mechanics Advisor: Keegan J. Moore .
2015 – 2017	M.Eng. in Thermal Power Engineering Zhejiang University, Hangzhou, Zhejiang, China Thesis: “Bearing Fault Diagnosis based on Improved Stochastic Resonance Method” Advisor: Wei Zhong .
2008 – 2013	B.S. in Mechanical Engineering Escuela Politécnica Nacional, Quito, Pichincha, Ecuador Thesis: “The Design of a Heat Exchanger for the Cupola Furnace” Advisor: Fausto Oviedo .

PROFESSIONAL EMPLOYMENT

2025 –	Postdoctoral Associate , Department of Applied Mathematics, University of Colorado Boulder, Boulder, Colorado, USA. Advisors: David Bortz , Stephen Becker .
2022 – 2025	Graduate Research Assistant – Fulbright Fellow , Department of Mechanical and Materials Engineering, University of Nebraska – Lincoln.
2021 – 2022	Graduate Teaching Assistant – Fulbright Fellow , Department of Mechanical and Materials Engineering, University of Nebraska – Lincoln.
2018 – 2021	Lecturer , Department of Basic Training, Escuela Politécnica Nacional.
2018	Lecturer , Department of Industrial Engineering, Universidad Tecnológica Indoamérica, Quito, Ecuador.
2015 – 2017	Graduate Research Assistant , Department of Energy, Zhejiang University.
2013 – 2014	Lecturer , School of Mechanical Engineering, Escuela Politécnica Nacional.

GRANT & FELLOWSHIPS

Awarded Research Grant

2026	Genesis: Transforming Science and Energy with AI. “ Weak-Form Agentic Digital Twins for Real-Time Control of Magnetically Confined Fusion Systems Under Sparse Observation ”. Awarded: \$750k. Role: Collaborator .
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Fellowships & Awards - University of Nebraska-Lincoln (Total: \$44k)

2025	5. Doctor Honoris Causa, Humanismus cum Rigoro Scientifico. Awarded by the Universidad GESTALT – Mexico; Ibarra – Ecuador.
2023	4. Travel Award, College of Engineering, University of Nebraska – Lincoln.
2022	3. Travel Award, College of Engineering, University of Nebraska – Lincoln.
2021 – 2024	2. Othmer Fellowship. University of Nebraska – Lincoln’s graduate studies for academic merit.
2021 – 2023	1. Fulbright Faculty Development Program to study a Ph.D. program in the United States.

Fellowships & Awards - Zhejiang University (Total: \$19k)

2017	2. Best Presenter of the session, IEEE CSPA Conference, Penang, Malaysia.
2014 – 2018	1. Chinese Government Scholarship. Language training and Master’s studies.

Fellowships & Awards - Escuela Politécnica Nacional (Total: \$7k)

2012 – 2013	2. Merit-based Teaching Assistantship. Casting Laboratory
2013	1. Academic Excellence. Top 2 % in the Mechanical Engineering program.

RESEARCH INTERESTS

Weak form Scientific Machine Learning, Data-driven modeling and identification of dynamical systems, Model predictive control, Theoretical and experimental nonlinear dynamics and vibrations, Signal processing, Rotating machinery fault diagnosis.

PUBLICATIONS

Peer-Reviewed Journals

2026	12. López C. , Partidge M., De Pascuale S., Lore J., Christlieb A.J., Becker S., Bortz D.M., “ WSINDy for Model Predictive Control with Applications to Fusion, Drones, and Chaos ”, <i>Proceedings of the Royal Society A</i> .
	11. Ghosh S., Andrulonis A., López C. , Singh A., Moore K.J., “ Energy-Based Dual-Phased Dynamics Identification for Forced Nonlinear Vibrations ”, <i>Mechanical Systems and Signal Processing</i> .
	10. López C. , Herzlieb J.E., Moore K.J., “ System Identification via Validation and Adaptation for Model Updating Applied to a Nonlinear Cantilever Beam ”, <i>ASME Journal of Computational and Nonlinear Dynamics</i> .
	9. López C. , Moore K.J., “ System Identification via Validation and Adaptation (SIVA) for Model Identification of Nonlinear Vibrating Structures ”, <i>Nonlinear Dynamics</i> .
2025	8. López C. , Naranjo Á., Salazar D., Moore K.J., “ Weak-form modified sparse identification of nonlinear dynamics ”, <i>Journal of Computational Physics</i> .
	7. López C. , Moore K.J., “ Energy-based dual-phase dynamics identification of clearance nonlinearities ”, <i>Nonlinear Dynamics</i> .
	6. López C. , Singh A., Naranjo Á., Moore K.J., “ A data-driven, energy-based approach for identifying governing dynamics in vibrating structures directly from measurements ”, <i>Mechanical Systems and Signal Processing</i> .
2024	5. López C. , Moore K.J., “ Enhanced adaptive linear chirplet transform for crossing frequency trajectories ”, <i>Journal of Sound and Vibration</i> .

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| 2022 | 4. López C. , Naranjo A., Lu S., Moore K.J., “Hidden Markov model based stochastic resonance and its application to bearing fault diagnosis”, <i>Journal of Sound and Vibration</i> . |
| | 3. López C. , Wang D., Naranjo A., Moore K.J., “Box-Cox-sparse-measures-based blind filtering: Understanding the difference between the maximum kurtosis deconvolution and the minimum entropy deconvolution”, <i>Mechanical Systems and Signal Processing</i> . |
| 2017 | 2. López C. , Zhong W., Lu S., Cong F., Cortese I., “Stochastic resonance in an underdamped system with FitzHug-Nagumo potential for weak signal detection”, <i>Journal of Sound and Vibration</i> . |
| | 1. Zhong W., Chen X., Zhou Y., Wu Y., López C. , “Optimization of a solar aided coal-fired combined heat and power plant based on changeable integrate mode under different solar irradiance”, <i>Solar Energy</i> . |

Preprint

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| S | Ghosh S., López C. , Singh A., Moore K.J., “Decomposition-based Energy-based Dual-Phase Dynamics Identification for Nonlinear MDOF Systems”, <i>Mechanical Systems and Signal Processing</i> . |
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CONFERENCE PRESENTATIONS & PROCEEDINGS

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| 2027 | 17. Ghosh S., López C. , Singh A., Moore K.J. “Energy-based Dual-phase Dynamics for Nonlinear System Identification of Multi-Degree-of-Freedom Systems”, <i>International Modal Analysis Conference (IMAC) XLV</i> , Kissimmee, FL, United States, February 1-4. |
| 2026 | 16. Ghosh S., López C. , Singh A., Moore K.J. “Energy-based Dual-Phase Dynamics Identification For Multiple-Degree-of-Freedom Nonlinear Structures”, <i>NODYCON – The Fifth International Nonlinear Dynamics Conference</i> , Rome, Italy, September 20–23. |
| | 15. López C. , Becker S., Bortz D.M. “Data-Driven MPC via WSINDy for Plasma, Drone Control, and Chaotic Systems”, <i>Society for Industrial and Applied Mathematics – Annual Meeting</i> , Cleveland, OH, United States, July 6–10. |
| | 14. López C. , Moore K.J. “Data-driven Structural System Identification via Validation and Adaptation”, <i>IMAC XLIV</i> , Palm Springs, CA, United States, January 19-22. |
| | 13. Ghosh S., Androlonis A., López C. , Moore K.J. “A Data-driven Method for System Identification of Forced Nonlinear Vibrations Using Energy Conservation”, <i>IMAC XLIV</i> , Palm Springs, CA, United States, January 19-22. |
| 2025 | 12. López C. , Moore K.J. “Generative Adversarial Networks with Embedded Physical Knowledge for Structural System Identification”, <i>International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE)</i> , Anaheim, CA, United States, August 17–20. |
| | 11. Ghosh S., Androlonis A., López C. , Moore K.J. “Data-driven Identification of Forced Nonlinear Vibrations using Conservation of Energy”, <i>IDETC/CIE</i> , Anaheim, CA, United States, August 17–20. |
| | 10. López C. , Singh A., Naranjo A., Moore K.J. “A Data-Driven Methodology to Identify the Governing Dynamics in Vibrating Structures Through the Analysis of Mechanical Energy”, <i>IMAC XLIII</i> , Orlando, FL, United States, February 10-13. |
| 2024 | 9. López C. , Singh A., Naranjo A., Moore K.J. “A data-driven methodology to identify the governing dynamics in vibrating structures through the analysis of mechanical energy”, <i>European Nonlinear Oscillations Conference</i> , Delft, Netherlands, July 22-26. |
| | 8. Oliveira F., Johnson B., Singh A., Moore K.J. “Mathematical Modeling of the Nonlinear Dynamics of Bolted Joint Loosening”. Presenter . <i>European Nonlinear Oscillations Conference 2024</i> , Delft, Netherlands, July 22-26. |

	7.	López C. , Ramsey T., Moore K.J. “Digital Engineering Test and Modeling Process: Autonomous Methods for Reconciling Test and Model Results”, <i>JANNAF 18th Modeling and Simulation</i> , Oklahoma City, OK, United States, May 6–9.
	6.	López C. , Moore K.J. “Enhanced Adaptive Linear Chirplet Transform for Multi-Component Signals with Intersecting Instantaneous Frequencies”, <i>IMAC XLII</i> , Orlando, FL, United States, January 29 – February 1.
	5.	Arroyo J., Ramsey T., López C. , Moore K.J. “Learning Missing Data in Measurements using Meta-Modeling Techniques”, <i>IMAC XLII</i> , Orlando, FL, United States, January 29 – February 1.
2023	4.	López C. , Singh A., Naranjo A., Moore K.J. “Data-driven Euler-Lagrange approach for Time Series Analysis”, <i>IMAC XLI</i> , Austin, TX, United States, February 13–16.
2022	3.	López C. , Naranjo A., Moore K.J. “Hidden Markov Model based Stochastic Resonance for bearing fault diagnosis”, <i>IMAC XL</i> , Orlando, FL, United States, February 7–10.
2017	2.	López C. , Zhong W., Cong F., Hidalgo V., “ Bearing fault diagnosis based on Ensemble Empirical Mode Decomposition and Teager energy operator ”, <i>IEEE 13th International CSPA</i> , Penang, Malaysia, March 10–12.
2016	1.	López C. , Zhong W., Zheng M., “ Short-term Electric Load Forecasting Based on Wavelet Neural Network, Particle Swarm Optimization and Ensemble Empirical Mode Decomposition ”, <i>Energy Procedia. 8th International Conference on Applied Energy</i> , Beijing, China, October 8–11.

WORKSHOPS

2026	4.	Scientific Machine Learning. University of Notre Dame, South Bend, IN, USA. September 26–27. Role: Attendee .
	3.	Society of Industrial and Applied Mathematics Annual Meeting. Weak-Form Scientific Machine Learning: Theory and Applications Minisymposium. Cleveland, OH, USA. July 6–10. Role: Co-organizer .
	2.	WSINDy for model predictive control with applications to plasma boundary dynamics, drone tracking trajectory, and chaotic systems. Postdoctoral seminar, University of Colorado Boulder, Boulder, CO, USA. April 29. Role: Presenter .
2025	1.	Center for Hierarchical and Robust Modeling of Non-Equilibrium Transport Annual Meeting. University of Washington, Seattle, WA, USA. December 1–3. Role: Presenter .

INDUSTRY EXPERIENCE

2022 – 2023		Cruze Distribution LLC dba ARYSE under AFOSR STTR Phase II, “STTR: Performance Knee Brace”. Finite element models were used to simulate and experimentally test components of a new type of knee brace.
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TEACHING EXPERIENCE & ACTIVITIES

University of Nebraska – Lincoln

Professional Development

2025	-	College of Engineering Graduate Student Teaching Fellow.
	-	Center for the Integration of Learning @ Nebraska Associate Level Certification.

Course Taught

2021 – | MENG 3250 Mechanics of Elastic Bodies (Teaching Assistant).
2022

Escuela Politécnica Nacional

Courses Taught

2018 – | - BAS030 Fundamentals of Physics
2021 - FIS145 Newtonian Mechanics: Practical Component and FIS151 Laboratory.
2013 – | - MEC1B2 Casting Technology
2014 - MEC4E3 Numerical Methods
- MEC4B4 Mechanics of Materials
- MEC7F4 Welding Production and Maintenance.
2012 – | MEC1B2 Casting Laboratory (Laboratory Assistant and Auxiliar).
2013

Universidad Tecnológica Indoamérica

Courses Taught

2018 | CAD/CAM/CIM and Maintenance.

Mentoring

2026 | 8. Tony Samour, University of Colorado Boulder, Master student.
2025 | 7. Jackson E. Herzlieb, Georgia Institute of Technology, Undergraduate student.
2024 | 6. Alexis Andrulonis, Georgia Institute of Technology, Undergraduate student.
5. Sayantan Ghosh, University of Nebraska – Lincoln, Ph.D. student.
2023 | 4. Thomas Ramsey, University of Nebraska – Lincoln, Undergraduate student.
2022 | 3. Javier Arroyo, University of Nebraska – Lincoln, Ph.D. student.
2014 | 2. Sebastián Camacho. “Design and simulation of a clay modeling lathe”, Escuela Politécnica Nacional, Undergraduate student, committee member.
1. Maritza Chuquirima and Lenin Cortez. “Study and Obtaining of Iron Metal from Ferruginous Sands”, Escuela Politécnica Nacional, Undergraduate students, committee member.

SERVICE

2024 | 4. Certified Peer Reviewer: Elsevier, Researcher Academy.
2023 | 3. Selection Committee – Fulbright Commission in Ecuador: Evaluation of applications.
2022 | 2. Intermediary for Academic Collaboration: Facilitated research partnership between University of Nebraska – Lincoln’s Mechanical & Materials Engineering Department and Escuela Politécnica Nacional’s Mechanical Engineering School.
2021 – | 1. Reviewer:

Journals

– Applied Mathematical Modelling	– Journal of Nondestructive Evaluation
– Chaos, Solitons and Fractals	– Journal of Signal, Image and Video Proc.
– Communications Physics	– Journal of Sound and Vibration
– Discover Applied Science	– Mechanical Systems and Signal Proc.
– IEEE Trans. on Industrial Electronics	– Nonlinear Dynamics
– IEEE Trans. on Signal Processing	– ROBOMECH Journal
– Journal of Aircraft	– Scientific Reports
– Structural Health Monitoring	

Conference

International Design Engineering Technical Conferences & Computers and Information in
Engineering Conference

LANGUAGES

Spanish (native), English (advanced), Mandarin (limited working proficiency).